

ALFAAL06358

## 2-Methylbutyric acid

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | 2-Methylbutyric acid<br>2-Methylbutyric acid                                                                                                                                                                                                                       |
| Cat No. :                     | L06358                                                                                                                                                                                                                                                             |
| CAS No                        | 116-53-0                                                                                                                                                                                                                                                           |
| Molecular Formula             | C5 H10 O2                                                                                                                                                                                                                                                          |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                       |                     |                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|
| Physical State<br>Liquid                                                                                                                              | Appearance<br>Clear | Odor<br>pungent |
| <b>Emergency Overview</b><br>Combustible liquid. Harmful if swallowed. Harmful in contact with skin. Causes severe skin burns and eye damage. Stench. |                     |                 |

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Flammable liquids.                | Category 4   |
| Acute Oral Toxicity               | Category 4   |
| Acute Dermal Toxicity             | Category 4   |
| Skin Corrosion/Irritation         | Category 1 B |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements



Signal Word

Danger

**Hazard Statements**

H227 - Combustible liquid

H314 - Causes severe skin burns and eye damage

H302 + H312 - Harmful if swallowed or in contact with skin

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P270 - Do not eat, drink or smoke when using this product

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Combustible material.

**Health Hazards**

Harmful if swallowed. Harmful in contact with skin. Corrosive. Causes skin and eye burns.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Stench. Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                | CAS No   | Weight % |
|--------------------------|----------|----------|
| Butanoic acid, 2-methyl- | 116-53-0 | 98       |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Call a physician immediately. If possible drink milk afterwards.

**Most important symptoms and effects**

Difficulty in breathing. Causes burns by all exposure routes. . Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Alcohol resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Combustible material. Flammable. Containers may explode when heated.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not let this chemical enter the environment. Remove all sources of ignition.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Keep away from open flames, hot surfaces and sources of ignition.

**Storage**

Keep refrigerated. Keep container tightly closed. Keep away from heat, sparks and flame. Corrosives area.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION**

## 2-Methylbutyric acid

## Control Parameters

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours

## Exposure Controls

## Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

**Appearance** Clear  
**Physical State** Liquid

## 2-Methylbutyric acid

|                                         |                                 |                                        |
|-----------------------------------------|---------------------------------|----------------------------------------|
| Odor                                    | pungent                         |                                        |
| Odor Threshold                          | No data available               |                                        |
| pH                                      | No information available        |                                        |
| Melting Point/Range                     | -70 °C / -94 °F                 |                                        |
| Softening Point                         | No data available               |                                        |
| Boiling Point/Range                     | 176 - 177 °C / 348.8 - 350.6 °F |                                        |
| Flash Point                             | 77 °C / 170.6 °F                | Method - No information available      |
| Evaporation Rate                        | No data available               |                                        |
| Flammability (solid,gas)                | Not applicable                  | Liquid                                 |
| Explosion Limits                        | Lower 1.2<br>Upper 5.7          |                                        |
| Vapor Pressure                          | 0.5 hPa @ 20 °C                 |                                        |
| Vapor Density                           | No data available               | (Air = 1.0)                            |
| Specific Gravity / Density              | 0.936                           |                                        |
| Bulk Density                            | Not applicable                  | Liquid                                 |
| Water Solubility                        | 45 g/L (20°C)                   |                                        |
| Solubility in other solvents            | No information available        |                                        |
| Partition Coefficient (n-octanol/water) |                                 |                                        |
| Component                               | log Pow                         |                                        |
| Butanoic acid, 2-methyl-                | 1.8                             |                                        |
| Autoignition Temperature                | 495 °C / 923 °F                 |                                        |
| Decomposition Temperature               | > 400°C                         |                                        |
| Viscosity                               | 2.1 mPa s at 20 °C              |                                        |
| Explosive Properties                    |                                 | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available        |                                        |
| Molecular Formula                       | C5 H10 O2                       |                                        |
| Molecular Weight                        | 102.13                          |                                        |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------|
| Stability                | Stable under normal conditions.                                                          |
| Hazardous Reactions      | No information available.                                                                |
| Hazardous Polymerization | No information available.                                                                |
| Conditions to Avoid      | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. |
| Materials to avoid       | Bases. Reducing Agent.                                                                   |

Hazardous Decomposition Products Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component                | LD50 Oral                 | LD50 Dermal                                                  | LC50 Inhalation |
|--------------------------|---------------------------|--------------------------------------------------------------|-----------------|
| Butanoic acid, 2-methyl- | LD50 = 1870 µL/kg ( Rat ) | LD50 = 1367 mg/kg ( Rabbit )<br>LD50 = 2228 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

## 2-Methylbutyric acid

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory</b>                                | No data available                                                                                                                                                                                                                                          |
| <b>Skin</b>                                       | No data available                                                                                                                                                                                                                                          |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                                                                                                                                          |
| <b>(f) carcinogenicity;</b>                       | No data available<br>There are no known carcinogenic chemicals in this product                                                                                                                                                                             |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                                                                                                                                          |
| <b>(h) STOT-single exposure;</b>                  | No data available                                                                                                                                                                                                                                          |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                                                                                                                                          |
| <b>Target Organs</b>                              | No information available.                                                                                                                                                                                                                                  |
| <b>(j) aspiration hazard;</b>                     | No data available                                                                                                                                                                                                                                          |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information                                                                                                                                          |
| <b>Symptoms / effects, both acute and delayed</b> | Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated |

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component                | Freshwater Fish                             | Water Flea | Freshwater Algae | Microtox |
|--------------------------|---------------------------------------------|------------|------------------|----------|
| Butanoic acid, 2-methyl- | LC50: > 1000 mg/L, 96h static (Danio rerio) |            |                  |          |

**Persistence and Degradability** Readily biodegradable  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component                | log Pow | Bioconcentration factor (BCF) |
|--------------------------|---------|-------------------------------|
| Butanoic acid, 2-methyl- | 1.8     | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**SAFETY DATA SHEET****2-Methylbutyric acid**

|                                            |                                                                                                                                                                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                        |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                     |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. |

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                                |                                           |
|--------------------------------|-------------------------------------------|
| <b>UN-No</b>                   | UN3265                                    |
| <b>Proper Shipping Name</b>    | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Butanoic acid, 2-methyl-                  |
| <b>Hazard Class</b>            | 8                                         |
| <b>Packing Group</b>           | II                                        |

**IMDG/IMO**

|                                |                                           |
|--------------------------------|-------------------------------------------|
| <b>UN-No</b>                   | UN3265                                    |
| <b>Proper Shipping Name</b>    | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Butanoic acid, 2-methyl-                  |
| <b>Hazard Class</b>            | 8                                         |
| <b>Packing Group</b>           | II                                        |

**IATA**

|                                |                                           |
|--------------------------------|-------------------------------------------|
| <b>UN-No</b>                   | UN3265                                    |
| <b>Proper Shipping Name</b>    | Corrosive liquid, acidic, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Butanoic acid, 2-methyl-                  |
| <b>Hazard Class</b>            | 8                                         |
| <b>Packing Group</b>           | II                                        |

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Special Precautions for User</b> | No special precautions required |
|-------------------------------------|---------------------------------|

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|--------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Butanoic acid, 2-methyl- | -                                                   | -                                       | X    | X     | 204-145-2 | X    | X   | X     | X    | X    | X    | KE-23544 |

**National Regulations****SECTION 16. OTHER INFORMATION**

## 2-Methylbutyric acid

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 07-Jun-2010  
**Revision Date** 22-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**